

Methods

Channel-Level Cortical Activation

To identify specific channels driving group differences in cortical activation, a channel-level LMM was estimated for each of the 15 HbO channels included in the ROI analyses (channel S4-D1 was excluded due to an insufficient number of subjects with valid data). For each channel, the model was:

$$\text{beta} \sim \text{group} * \text{sessao_num} + (1 \mid \text{subject})$$

where group (reference: nao_acelerado) and sessao_num (integer, 0-9) were fixed effects, and a random intercept per subject was included. Models were estimated using maximum likelihood (ML) with the BFGS optimiser. The critical term was the group x session interaction coefficient, representing the differential rate of HbO change per session in the Accelerated relative to the Non-Accelerated group. Multiple comparison correction used the Benjamini-Hochberg false discovery rate (FDR) procedure ($\alpha = 0.05$) across all 15 channel-level tests. Anatomical localisation followed the standard 10-10 electrode system with probabilistic region assignments.

Correlation Between HbO Slope and Reading Speed Slope

For the six FDR-significant channels, individual HbO slopes were computed as the OLS regression coefficient of $\text{beta} \sim \text{sessao_num}$ per participant per channel (training sessions 1-9). Individual reading speed slopes were computed as the OLS coefficient of $\text{TimeRm} \sim \text{sessao_num}$ per participant. Spearman correlations were computed between HbO slope and TimeRm slope for each channel across three scopes: all subjects ($N = 13$; SUBJ_030 excluded as a previously identified outlier), Accelerated only ($N = 7$), and Non-Accelerated only ($N = 6$). FDR correction (Benjamini-Hochberg) was applied within each scope across the six channel tests.

Results

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Six of the 15 channels tested reached FDR significance ($q < .05$) for the group x session interaction (Table 5). In the Frontal ROI, one channel was significant: S3-D1 (L Middle Frontal Gyrus, 66%; 10-10: AF3-F5), $\text{beta} = -0.296$, $p = .003$, $q = .014$, 95% CI [-0.490, -0.103]. The negative coefficient indicates that the Accelerated group showed a steeper decline in HbO activation per session relative to the Non-Accelerated group.

In the Temporal ROI, five channels were FDR-significant. Four showed negative interaction coefficients (steeper HbO decline in the Accelerated group): S5-D4 (L Middle Temporal Gyrus, 41%; P7-TP7), $\text{beta} = -0.297$, $p = .002$, $q = .014$, 95% CI [-0.481, -0.112]; S7-D7 (L Middle Temporal Gyrus, 55%; T7-C5), $\text{beta} = -0.298$, $p = .002$, $q =$

.014, 95% CI [-0.490, -0.106]; S5-D6 (L Middle Temporal Gyrus, 47%; P7-P5), beta = -0.318, p = .006, q = .021, 95% CI [-0.547, -0.090]; and S7-D4 (L Middle Temporal Gyrus, 72%; CP5-TP7), beta = -0.197, p = .009, q = .023, 95% CI [-0.346, -0.049]. One channel showed a positive interaction: S5-D5 (L Inferior Temporal Gyrus, 29%; P7-P9), beta = +0.215, p = .007, q = .021, 95% CI [0.059, 0.372].

Table 5. Channel-Level LMM — FDR-Significant Channels (HbO, Group x Session)

Channel	10-10	Region	ROI	Beta	p	q(FDR)	95% CI
S3-D1	AF3-F5	L Middle Frontal Gyrus (66%)	Frontal	-0.296	.003	.014	[-0.490, -0.103]
S5-D4	P7-TP7	L Middle Temporal Gyrus (41%)	Temporal	-0.297	.002	.014	[-0.481, -0.112]
S7-D7	T7-C5	L Middle Temporal Gyrus (55%)	Temporal	-0.298	.002	.014	[-0.490, -0.106]
S5-D6	P7-P5	L Middle Temporal Gyrus (47%)	Temporal	-0.318	.006	.021	[-0.547, -0.090]
S5-D5	P7-P9	L Inferior Temporal Gyrus (29%)	Temporal	+0.215	.007	.021	[0.059, 0.372]
S7-D4	CP5-TP7	L Middle Temporal Gyrus (72%)	Temporal	-0.197	.009	.023	[-0.346, -0.049]

FDR: Benjamini-Hochberg; 15 tests; ML estimation; BFGS.

Correlation Between HbO Slope and Reading Speed Slope

Spearman correlations between individual HbO slopes and TimeRm slopes were computed for each of the six FDR-significant channels. No correlation reached significance after FDR correction in any scope. In the full-sample analysis (N = 13), rho values ranged from -0.456 (S7-D4, p = .117) to -0.148 (S5-D4, p = .629). In the Non-Accelerated subgroup (N = 6), S7-D4 showed a nominally significant negative correlation (rho = -0.886, p = .019), which did not survive FDR correction (q = .113). All Accelerated subgroup correlations were non-significant (all p > .29).